

CHEOL KIM (김철)

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RESEARCH INTERESTS

Physical AI · Agentic AI · Large Language Models · Information Retrieval

EDUCATION

Sungkyunkwan University (SKKU)

March 2022 – August 2027 (expected)

B.S. in Computer Science and Engineering, College of Computing and Informatics

- Selected coursework: Computer Vision, Fundamentals of Machine Learning, Linear Algebra, Deep Learning

McLean High School · Virginia, United States

August 2018 – June 2021

RESEARCH EXPERIENCE

DIAL Lab, Sungkyunkwan University

August 2025 – August 2026

Undergraduate Research Intern, advised by Prof. Jongwuk Lee

Ambiguity Resolution for Text-to-SQL (Academic Research)

March 2026 – June 2026

With a first-year Ph.D. student; manuscript prepared for VLDB submission.

- Co-designed a training-free method for ambiguous natural-language questions: enumerate candidate interpretations, prune them against database contents, ground unambiguous spans, then expand only the underspecified ones.
- Improved recall by **11.2%** over training-free zero-shot baselines on AMBIQT, while returning an average of only 5 candidate queries.
- Evaluated across five Text-to-SQL benchmarks including BIRD and Spider; reproduced 8 baseline systems for comparison.
- Co-authored the manuscript, including the abstract and implementation details.

Text-to-SQL for Manufacturing Databases (Industry Research)

September 2025 – March 2026

Collaboration with Wezon on a natural-language interface for a production database.

- Proposed diversity and coverage metrics over question embeddings, shaping the pairwise cosine-similarity distribution to detect and correct sampling bias.
- Built a 5,000-pair question–SQL dataset with a graduate student, combining programmatic generation with LLM-based paraphrasing, and validated the generated pairs by hand.
- Benchmarked open-source LLMs (Qwen2.5, Llama, Codestral) and fine-tuned Qwen2.5; delivered a model-agnostic few-shot pipeline so the client could substitute commercial APIs without retraining.

INDUSTRY EXPERIENCE

Samsung SDS

June 2026 – August 2026

Summer Intern, Architecture Group, ERP Technical Team

- Selected as one of five interns company-wide through nationwide open recruitment.
- Completed a two-week SAP ERP integration consulting assignment and presented the results to department leadership.

TECHNICAL SKILLS

Languages: Python, C **Frameworks:** PyTorch, Hugging Face Transformers, vLLM

Tools: Git, Linux, Weights & Biases

TEACHING EXPERIENCE

Teaching Assistant · Freshmen Programming Preparatory Course, SKKU

February 2025

- Advised student-led team projects in a two-week Python course; the supervised team was one of five recognized among 28 by TA evaluation and student vote.

Instructor, Introductory Programming · Suwon Christian International School

March 2025 – July 2025

- Taught a semester-long Python and computing course to middle and high school student groups.

LANGUAGES

Korean (Native) · English (Fluent) — TOEIC 985, 2025; TOEFL 113, 2021